

# The Ten Scenarios

You will be assigned **one** scenario. You work it over **two class days**: Day 1 you take it apart, Day 2 you decide and defend. Everything is on this packet.

| #  | SCENARIO                                                                                                                              |
|----|---------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <b>The design that only works in the lab</b><br>Your prototype passes every bench test but fails the first time it is used outdoors.  |
| 2  | <b>Cheaper or safer</b><br>A different material cuts the cost by 30% but has not been tested for the loads your design will see.      |
| 3  | <b>The client wants it changed</b><br>Your client asks for a change that makes the product look better but weaker.                    |
| 4  | <b>Data that does not agree</b><br>Two of your five test trials give results far outside the other three.                             |
| 5  | <b>Someone else's idea</b><br>A teammate's concept is stronger than yours. Your name is on the project either way.                    |
| 6  | <b>The deadline moved up</b><br>You lose two weeks of testing. You can still deliver, but with less evidence that it works.           |
| 7  | <b>Accessibility was not considered</b><br>Your design works well for most users but cannot be operated one-handed.                   |
| 8  | <b>The failure nobody saw</b><br>Your product works for six months, then fails. You cannot reproduce the failure.                     |
| 9  | <b>Scaling up</b><br>Your design works as one unit. Making a thousand requires changes you did not plan for.                          |
| 10 | <b>Who is responsible</b><br>A component you bought fails and causes your product to fail. The supplier says it was used incorrectly. |

# Take It Apart

Name \_\_\_\_\_ Scenario # \_\_\_\_\_ Date \_\_\_\_\_

## Bell work — first 10 minutes

**Your design passes every test you ran, and then fails for a real user. What does that tell you about your testing?**

**Format:** It tells me my testing \_\_\_\_\_, because \_\_\_\_\_.

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## 1 Say the problem in one sentence

Not the story — the actual problem that has to be solved.

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## 2 Who is affected, and how?

| WHO   | WHAT IT COSTS THEM |
|-------|--------------------|
| _____ | _____              |
| _____ | _____              |
| _____ | _____              |

## 3 What do you still need to know?

Three questions you would have to answer before deciding anything.

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## 4 What is the real pressure here?

Money, time, a person, a reputation — name it.

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# Decide and Defend

Name \_\_\_\_\_ Scenario # \_\_\_\_\_ Date \_\_\_\_\_

## Bell work — first 10 minutes

Is a design decision still good if it turns out badly? Explain.

Format: \_\_\_\_\_, because a good decision is judged on \_\_\_\_\_, not on \_\_\_\_\_.

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## 5 Three options you could take

| OPTION      | WHAT IT COSTS |
|-------------|---------------|
| A.<br>_____ | _____         |
| B.<br>_____ | _____         |
| C.<br>_____ | _____         |

## 6 Your call — circle one: A B C

Three reasons, tied to the facts of your scenario, not to how you feel.

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## 7 What would have prevented this in the first place?

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## 60-second report out

Be ready to say: the problem in one sentence, your call, and your strongest reason. The class gets one question.